KiraOrder Final Solution Report

Falcons | SDC Hackathon 2026 | Version 2 | 31 October 2026

KiraOrder converts mixed-language wholesale messages into checked order drafts and delivery orders approved by a person. Its shared record connects interpretation, correction and decisions across clerk, owner and warehouse roles. Deployed frontend and backend components have evidence for selected approval, stock and access-control paths. This report presents the product, a documented order journey, the changes since the preliminary submission and the results available so far; business impact remains to be measured.

1 Team and Project Overview

Item	Project details
Project and team	KiraOrder for Orbyte, developed by Falcons
Challenge	Domain 2 Orbyte Smart SME Challenge
Institution	Xiamen University Malaysia
SDG focus	SDG 8.2 and 8.3: productivity and support for SME activity
Team responsibilities	Backend and deployment; AI integration; frontend; data and evaluation; baseline measurement and submission materials

Interactive workspace | Source code and README

The workspace opens in Local Demo without credentials. Real API access uses a privately arranged, role-specific session coordinated through the team portal.

2 Project Description

Malaysian wholesale SMEs receive orders that mix Malay, English and Chinese, product nicknames, quantities and packaging terms. A clerk must translate each message into catalogue items, verify prices and stock, resolve conditions and prepare a delivery order. Ambiguity creates additional clarification and makes handoffs between sales, the owner and the warehouse difficult to track.

KiraOrder turns a pasted message into a reviewable order draft. Exact customer aliases can resolve known items; the configured language model extracts product-family candidates, quantities, units, pack sizes and conditional phrases. Backend rules resolve catalogue matches, calculate prices and check stock. The interface retains source evidence and makes unresolved items visible. Clerks can clarify or correct the draft, while price exceptions require an owner decision.

A saved, eligible draft needs explicit human approval before a delivery order is generated and stock deducted. The practical value is a shared order record connecting the original message, corrections, prices and decisions. Clerks can review exceptions, owners control price decisions and warehouse staff receive restricted picking details. Rejection records a reason and closes the draft.

KiraOrder is intended as an order intake and approval module for Orbyte. The prototype uses pasted messages and copied replies; direct Orbyte integration and automatic WhatsApp messaging remain future work. Demonstrations and evaluation use synthetic business data. [E1, E2]

3 Technical Architecture and Implementation

Final System Architecture

A React and TypeScript application on GitHub Pages presents the order workspace. FastAPI on Replit handles authenticated business operations and persists workflow records in Neon PostgreSQL 17. Its configured extraction adapter calls Qwen2.5-7B-Instruct-AWQ through vLLM on RunPod. The model proposes structured interpretations; the backend determines catalogue resolution, prices, stock eligibility and permitted actions. [E1, E3, E6]

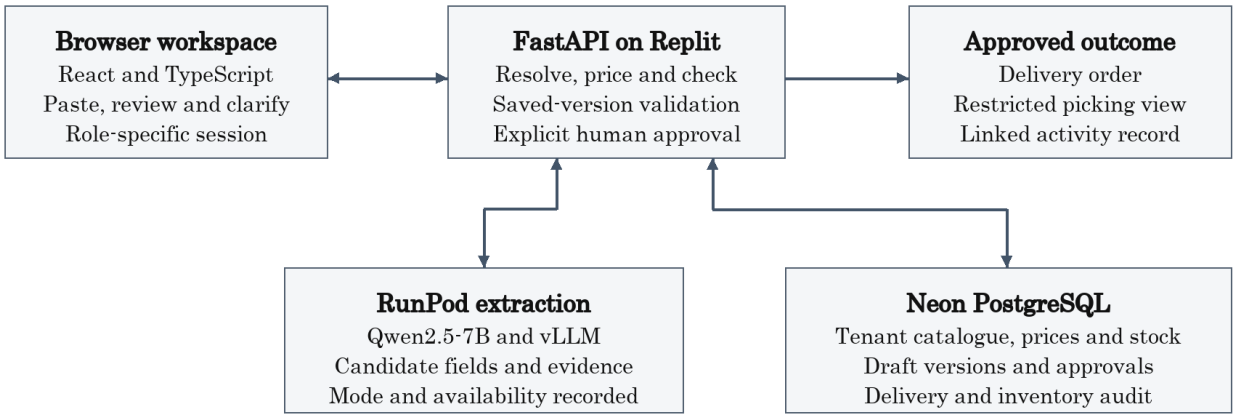


Figure 1. Real API architecture. Browser-only Local Demo is a separate execution mode.

Data and decision flow

The clerk selects a customer and supplies an order message. Catalogue and alias matching, together with the configured extraction path, produce candidate lines and conditions. The resolver checks tenant data and preserves unresolved questions. Saving creates a versioned draft; the user reviews the resulting prices, issues and allowed actions. Approval operates on the saved version. An eligible approval creates the delivery order, inventory deduction and audit events within the business transaction.

Human control and access

CLERK sessions prepare and approve eligible routine orders. OWNER sessions decide escalated requests, including price exceptions; an owner decision cannot bypass unresolved stock or product issues. WAREHOUSE sessions receive only permitted picking information. Identity and tenant scope come from the backend session. The warehouse projection excludes prices, customer contact details and the original order message. [E1, E4, E5]

Uncertainty and recovery

Ambiguous products, missing units and insufficient stock block approval until resolved. Manual recovery records the chosen item and its human origin, then reruns business checks. Pending and terminal records restrict further changes. Real API parsing can use LLM, FALLBACK, MIXED or ALIAS_ONLY paths, so the mode must be recorded for each evaluation. Local Demo runs catalogue rules in the browser and establishes neither live model quality nor backend persistence. Tokens used by real sessions remain in tab memory. [E1, E3, E6]

Order Walkthrough

In the documented C02 run, a clerk entered "Milo 2 bag" without a pack size. KiraOrder presented a choice, retained the human correction and generated a delivery order after saving and approval. This synthetic example was recorded on 30 October using frontend v1.4.0 and a reported real API connection in FALLBACK mode. [E13]

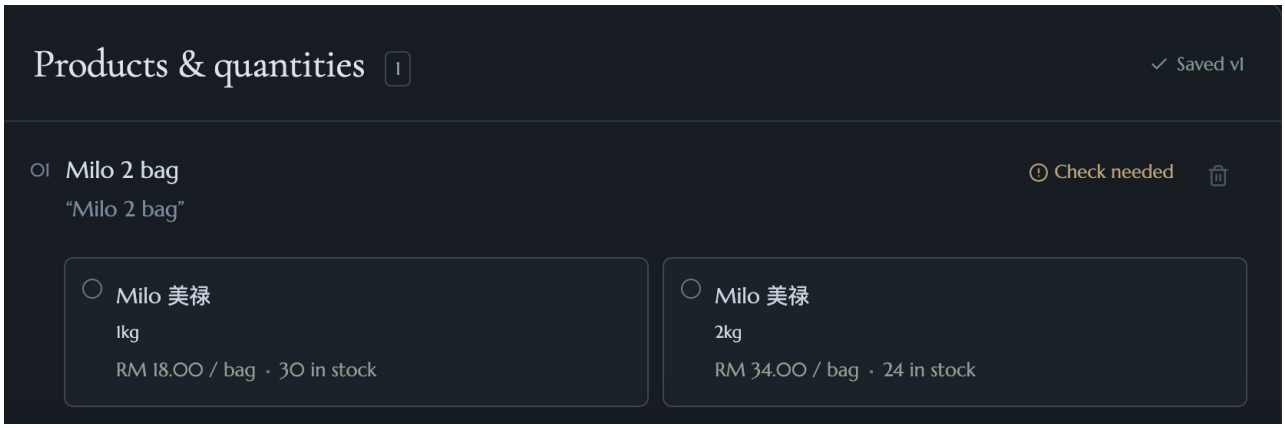


Figure 2. The missing pack size leaves both candidates unselected and approval blocked.

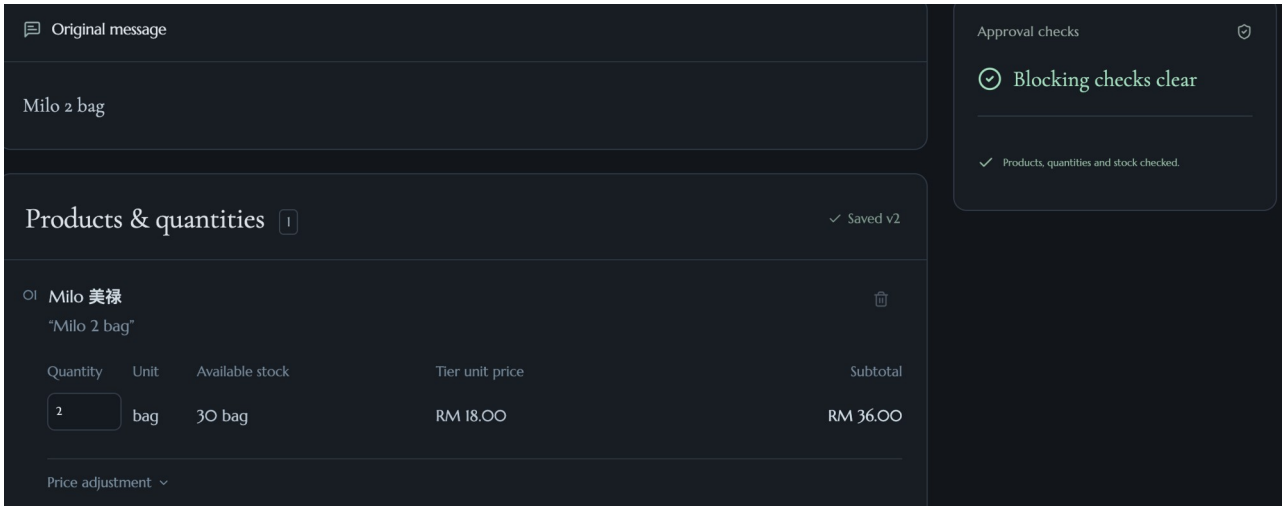


Figure 3. The clerk selected 1kg and saved v2. Checks cleared at 2 bag x RM18.00.

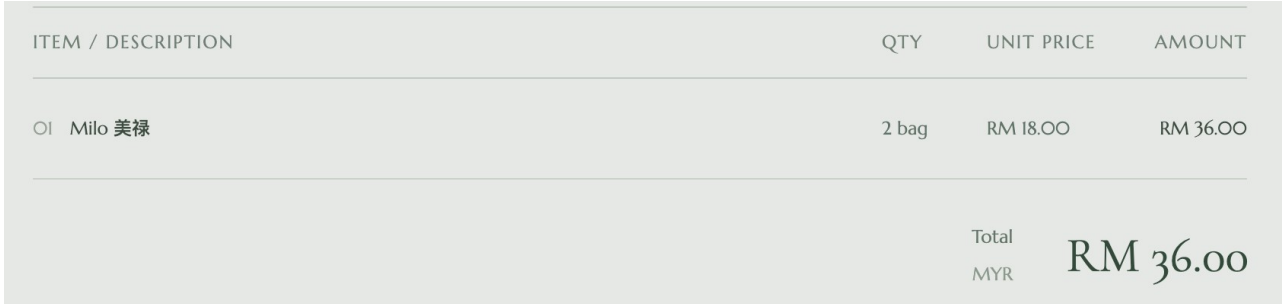


Figure 4. Approval produced a delivery order for RM36.00, tied to saved version 2.

These are cropped historical interface screenshots; unrelated browser content, contact details and record identifiers are excluded. The audit confirms SKU_MILO_1KG and MANUAL_SELECTED. The DO item label omits the pack size, a recorded presentation issue. C02 is manual UI evidence, not an LLM accuracy or database stock-delta test. [E13]

Implementation and Deliverables

The preliminary submission described the order workflow and a separately exercised model endpoint, with integrated validation unfinished. We then added recovery controls, clearer decision records and scoped evidence around that workflow. The comparison below connects the changes to their practical use. [E2]

Area	Preliminary position	Final change and user value
Workflow	Browser workflow; integrated validation incomplete	Role-bound API connection and saved drafts make the clerk's review state explicit. [E1, E6]
Recovery	Ambiguity visible; workflow validation unfinished	Catalogue correction and manual line entry retain human provenance and rerun business checks. [E6]
Decisions	Human approval and restricted picking designed	Owner reasons, order activity and delivery-order links expose the decision and its outcome. [E1, E12]
Evidence	Model calls recorded; accuracy unscored and workflow tests pending	A03 adds independent scoring; C07 checks competing approvals and stock within defined scenarios. [E3, E4]
Usability	Browser prototype; first-use baseline unmeasured	v1.5.2 improves workbench readability and responsive layouts, and refreshes Demo stock after other approvals. [E7]

Table 1. Evolution from the preliminary submission to the final build.

The delivery includes frontend v1.5.2, backend B03, operating instructions, source code and archived evaluation evidence. Release revisions are listed in the appendix. The final PRD, technical declaration and presentation materials require reconciliation. Dates follow the records; venue-specific completion has not been independently established. [E6, E7, E10]

4 Impact Verification

SDG achievement and business outcomes

KiraOrder targets SDG 8.2 through support for clerical productivity and SDG 8.3 through more reliable SME order records. Its demonstrated contribution is a controlled workflow; productivity and business-error outcomes require a field pilot. [E2, E11]

Measure	Preliminary target	Position at 31 October
Handling time	At least 50% lower median time against a validated manual baseline	Not measured. The initial 210-second baseline was a planning estimate; no valid paired user-trial result is available.
Silent mismatches	Zero observed silent product or quantity mismatches in the evaluated sample	Product and unit uncertainty was explicitly flagged in A03. Quantity and condition errors remain; the combined end-to-end target is not verified.
Price conditions	Human handling for all detected price conditions	Escalation and owner decisions are implemented and tested in selected cases. Condition-span errors remain; full detection recall is not established.
First-use completion	At least 80% completion without guidance	Not measured. Four planned F05 participant slots were cancelled and not run.

Table 2. Preliminary targets and current measurement status. [E2, E3, E8]

Verification method

We distinguish model extraction, business workflow checks, interface checks and independent user trials. The A03 model results use a 23-case development set with an independent answer key; the public archive contains a redacted scoring summary. Business tests inspect defined API responses, stored outcomes or browser observations. Each record retains its date, mode, tested version and limits. These samples do not constitute a representative study of SME users.

Socioeconomic value

The intended socioeconomic benefit is less repeated copying and clarification, visible responsibility for exceptions and appropriate information for warehouse staff. The preliminary estimate of up to 30% transcription or stockout discrepancies remains unvalidated; the proposed pilot below is needed to measure actual gains. [E2]

Deployment and commercial feasibility

The deployed components and synthetic tenants support review under defined conditions. GPU availability and cold starts affect live operation; Local Demo supplies a separate browser-based demonstration when a model session is unavailable. A pilot should establish support effort, operating cost and integration requirements before commercial rollout.

Proof and Evaluation Results

The 30 October A03 run directly exercised the production adapter in one warm-worker session. All 23 LLM calls completed without fallback, with no reported request duration above the configured 4.1-second threshold. Independent scoring measures extraction quality separately below. [E3]

A03 measure	Result	Interpretation
LLM call completion	23/23	Execution success; not extraction accuracy
Strict six-dimension case pass	8/23 (34.8%)	All required dimensions correct within a case
Product family	21/27 (77.8%)	Field-level result on 27 scored lines
Quantity	26/27 (96.3%)	Field-level result on the same scored lines
Unit of measure	19/27 (70.4%)	Seven known unit gaps remain
Condition type	8/8	Type scoring; does not establish full recall
Exact condition span	4/8	Four source-span truncation errors
Matching line count	20/23	Case-level structure check

Table 3. A03 development-set results, with separate denominators for each measure.

Model Limitations and Workflow Evidence

Nine UNKNOWN lines were explicitly flagged. Seven unit gaps entered backend clarification or blocking paths, but four condition-span truncations were not flagged at that extraction stage. The six canonical samples separately recorded five matching line-level results and one known condition-span limitation; an additional ignored-span merging difference was noted. Family was excluded from that six-sample scoring. Later isolated prompt experiments do not replace this published baseline or establish a deployed quality improvement. [E3]

Business integrity

The C07 online record covers six approval and inventory scenarios with 34 accepted-run HTTPS requests. Four concurrent routine approvals produced one successful approval and three already-approved responses, with one delivery order. Competing for the last stock unit produced one success and one insufficient-stock rejection. A prior timing attempt that failed to prove overlap was retained, followed by a successful targeted rerun. Fault rollback was tested separately in isolation. [E4]

The C08 warehouse API matrix passed 41/41 checks covering identity, access, method restrictions and tenant scope. Its projection contained four root fields and five fields per item, excluding the scanned price, contact and original-message fields. Coverage is limited to the tested warehouse boundary. [E5]

Additional workflow evidence

Workflow	Recorded result	Scope
Owner rejection	A below-tier request was rejected with a reason; the draft closed and the UI showed no generated delivery order. [E12]	Manual C05 browser evidence; no direct database readback
Manual recovery	Four isolated T1/T2 clerk and owner paths completed catalogue selection, zero-line recovery, approval and warehouse retrieval. [E6]	Synthetic provider and local PostgreSQL; no GPU call
Second tenant	The F05 record includes a T2 owner approval and warehouse reading, with observed stock changing from 38 to 36. [E8]	Selected real API operations in team-held records; no user trial

Engineering and remaining evaluation

The v1.5.2 release records 82/82 frontend tests, a successful build, resource-hash checks and layouts checked at 320, 768 and 1440 pixels. Formal TEST32 workflow execution remains NOT_RUN in B04: 12 logical cases and 14 executions. B05 user trials were not run; F05 completed a reduced engineering scope with exceptions and cancelled first-use trials. Overall product acceptance and user impact remain unverified. [E7, E8]

Proposed Pilot Measurement

A supervised pilot should compare equivalent manual and assisted orders with task order counterbalanced. Timing should start when the message is opened and end at a correct, approved delivery order, including clarification and correction. The protocol should fix sample size, timeout and error definitions before recruitment, report failures separately, and calculate time reduction as (manual median minus assisted median) divided by manual median. First-use completion should be measured without guidance. Silent mismatches and missed price conditions need separate counts. This is a proposed protocol; no pilot results are reported.

5 Market and Scalability

The initial users are small wholesalers that receive orders through chat, maintain product and price lists, and need owner approval for exceptions. KiraOrder organizes the work between receipt of a message and an approved delivery order, with less repeated transcription as the intended benefit.

An Orbyte module would connect customer, catalogue, inventory and accounting systems through agreed interfaces. T1 and T2 synthetic tenants already exercise separate catalogues and role contexts. Selected cross-tenant rejection and second-tenant workflow records support reuse within the tested cases. Commercial adoption and performance at scale remain untested. [E5, E8]

Onboarding would import the catalogue, map units and aliases, configure approval rules, validate representative orders and run a supervised pilot. Cost assessment should record hosting, database storage, GPU active time and support, then divide total operating cost by processed orders. Monthly volume, cold starts and correction effort would determine whether the service is economical. A subscription with an implementation fee is a possible model; pricing and willingness to pay are unvalidated.

The next priorities are the pilot described in Section 4, unit and specification improvements, repeatable model evaluation and business-system integration. Expansion would follow measured handling-time and control results, with explicit acceptance criteria before adding another tenant.

6 Eligibility and Readiness Checklist

The checklist records delivery status. Final portal submission requires a separate receipt.

Readiness item	Status	Basis or remaining action
Deployed prototype	Recorded	Pages v1.5.2 and a separately recorded Replit backend; verify the intended review session.
Installation and operation	Available	README and operating handoff explain setup, modes and role access.
Development history	Available	Incremental source and review history are retained in the repository.
Business validation	Partial	Selected approval, inventory, access and recovery paths have dated evidence.
Formal evaluation and UAT	Incomplete	Formal workflow tests and independent user-impact trials remain outstanding.
Final materials and template	Pending	Reconcile the final report, updated PRD, declaration, slides and official template.
Declarations and submission	Pending	Team confirmation, signature, final access check and final portal receipt are still required.

Existing submission records cover preliminary materials. [E10]

7 IP and Originality Declaration

Pre Existing and Prior Owned Work

The preliminary proposal declared no inclusion of team members' pre-existing work. The final declaration requires confirmation of that position and any subsequently added work. [E2]

Third Party Code Data and Models

The project uses third-party software, models, development tools and cloud services. Representative dependencies and recorded licences are listed below. The technical declaration remains the detailed resource inventory; it must be updated for the final dependency set, AI-assisted development tools, fonts and visual assets. [E9, E10]

Resource	Use	Licence or declaration position
Qwen2.5-7B-Instruct-AWQ and vLLM	Extraction and model serving	Apache 2.0, as recorded in the preliminary declaration
React, Vite, FastAPI and associated libraries	Application framework and tooling	Primarily MIT; consult the itemized declaration
TypeScript and project repository	Language and project code	Apache 2.0
PostgreSQL and psycopg	Storage and database access	PostgreSQL License and LGPL 3.0 respectively
pglast	SQL parsing utility	GPL 3.0 or later, as recorded in the declaration
GSAP, interface assets and fonts	Motion and presentation	Reconcile current packages and asset-specific terms in the final declaration
Replit, RunPod and Neon	Cloud services	Service terms and actual platform use must be disclosed

Evaluation uses team-created synthetic business data. This report excludes private credentials, customer contacts and private answer keys. Cropped interface evidence retains only the relevant synthetic workflow.

Employment Related IP Conflicts

Each member must confirm whether employment, internship or consulting terms create an ownership claim. Final confirmation is pending.

8 Declaration

By signing, the team leader confirms that the final submission accurately describes the work, third-party resources and evidence, and that all team declarations have been reviewed. Results apply to the dates, versions, samples and conditions stated.

Team leader name: ZHOU JINGXI



Signature

Date signed: 1.11.2026

Appendix Evidence and Version Register

The evidence register links claims to dated records. Repository sources are pinned to archive 2efa333; tested releases may be older, as noted below. Full private screenshots and answer keys are excluded.

E1 Product workflow and review instructions

README and product context. Describes roles, operating modes, manual input and copied replies.

E2 Signed preliminary proposal

25 October, pages 3 to 5. Initial workflow, integration and evaluation status, estimated baseline and impact targets.

E3 A03 independent model review

30 October. Single warm adapter run and independent redacted scoring. Application 555f0fa; AI module db6f0ce. Not a formal TEST32 or end-to-end workflow score.

E4 C07 online approval and stock review

31 October. Six online scenarios, 34 accepted-run requests. Replit 9cff61ed, application 9dfee33. Rollback evidence is isolated, and the failed timing attempt is retained.

E5 C08 warehouse access review

30 October. 41/41 API checks and field projection review. The record does not cover all clerk-to-owner access paths or a complete T2 business journey.

E6 B03 manual recovery delivery

31 October. Backend application 99ee169, Replit 6d6bafd8. Includes recovery, CI, release checks and the limits of independent review. Frontend behavior is separately implemented.

E7 Frontend v1.5.2 stock correction

31 October team release record: application 2efa333, Pages 0905ccf, 82/82 frontend tests. Public release metadata is linked below; it can change after this report.

E8 Evaluation scope and unperformed trials

B05 user trials NOT_RUN; B04 formal workflow tests unexecuted; F05 first-use trials cancelled and engineering scope reduced.

E9 Final frontend dependency manifest

v1.5.2 dependencies, including GSAP and the React interface stack. The manifest alone is not a complete licence or asset-clearance review.

E10 Submission materials and technical declaration

Archived preliminary proposal, technical declaration, slides, video and platform-use proofs. Final-round packaging requires an updated reconciliation.

E11 United Nations Sustainable Development Goal 8

Targets 8.2 and 8.3 provide the productivity and SME policy context for the intended impact.

E12 C05 owner rejection browser record

31 October. Rejection, terminal draft state and activity reviewed in historical FALLBACK mode; no database-wide verification.

E13 C02 pack size confirmation and delivery order

30 October, frontend v1.4.0, reported real API with FALLBACK. Manual UI sequence; independent review on 31 October. Figures 2 to 4 are cropped derivatives of hash-verified local originals.

B04 formal workflow scope | Public frontend release metadata